Douban Movie Short Review Sentiment Analysis Platform - Project Report

1. Cover Information

- **Project Name**: Douban Movie Short Review Sentiment Analysis Platform
- **Project Type**: Web Application Based on Web Scraping Data Services
- **Technical Direction**: Data Crawling, Text Cleaning, Sentiment Analysis, Visualization
- **Author Information**: Dengruihan / 210110067 / Odyssey
- **Submission Date**: 2026-04-13
- **GitHub Repository**: https://github.com/dengruihan/Douban_Film_Comment-monitoring
- **Online Deployment**: http://47.98.161.252:8501

2. Project Abstract

This project develops a movie short review sentiment analysis platform based on Python and Streamlit to address the practical problem of "how to quickly understand public sentiment about a movie." After users input a movie name, the system automatically searches for the target movie, crawls publicly available review data from Douban's mobile interface, performs text cleaning and sentiment analysis, and displays results through charts, word clouds, and tables.

The project implements a complete workflow from "external data acquisition" to "analysis result presentation," with clear data sources, reproducible technical solutions, and a visualization interface, meeting the course requirements for "building practical services based on network data."

3. Project Background and Problem Definition

In movie viewing decisions, film evaluation research, and public opinion observation, users typically need to read a large number of short reviews to judge a movie's reputation. However, directly browsing raw reviews presents the following challenges:

- Large volume of reviews with high manual screening costs;

- Dense information on original pages making it difficult to quickly extract overall trends;
- Noisy text in reviews that is not conducive to statistical analysis;
- Simple display of raw data does not reflect the value of "data services."

Therefore, the goal of this project is to build a public opinion analysis platform for ordinary users. By crawling publicly available movie short reviews and processing them, the system provides users with:

- Overall emotional distribution of movies;
- Time trend of review popularity;
- High-frequency keywords and core discussion content;
- Comparison capabilities between similar movies.

4. Data Sources and Collection Methods

4.1 Data Sources

This project uses Douban's mobile public interfaces as data sources, mainly including:

- Search API: `m.douban.com/rexxar/api/v2/search/movie`
- Movie Details API: `m.douban.com/rexxar/api/v2/movie/{id}`
- Reviews API: `m.douban.com/rexxar/api/v2/movie/{id}/interests`

The crawled fields mainly include:

- Movie Information: Movie ID, Movie Name, Average Rating, Category Information
- Review Information: Username, Star Rating, Review Text, Like Count, Review Time

4.2 Collection Workflow

1. User inputs movie name.
2. System calls the search API to get candidate movie list.
3. After user selects target movie, system reads the movie ID.
4. System crawls movie details and review data.
5. Data is written to local SQLite database.
6. Frontend reads database and completes analysis and display.

4.3 Data Collection Strategy

To avoid excessive pressure on the target website and reduce anti-crawling risks, the project adopts the following strategies:

- Default limit of 10 pages for collection;
- Fixed 20 reviews per page;
- Random delays between requests;
- Prioritize mobile interfaces to reduce instability caused by webpage structure changes;
- Add retry mechanisms for abnormal requests and failures.

5. Compliance, Ethics, and Security Statement

5.1 Compliance Considerations

This project is for course learning and technical demonstration purposes only. Data sources are publicly accessible content. The system only crawls single movie data when users actively initiate queries, without large-scale batch crawling.

Current version has implemented:

- Request frequency control;
- Optional Cookie environment variable configuration;
- No mandatory dependency on login status in code;
- Sensitive configuration managed through environment variables.

If deployed publicly in the future, the following should be added:

- robots.txt checking and access permission confirmation;
- Stricter rate limiting;
- Access logs and anomaly alerts;
- Manual confirmation of third-party platform service terms.

5.2 Ethics Statement

- Only publicly available review fields are collected, no private sensitive information is gathered;
- No user profiling is performed;
- Results are displayed mainly through aggregated analysis, not used for commercial secondary purposes;

- The system's analysis results are for reference only and do not represent official platform evaluation conclusions.

5.3 Security Statement

- API Keys, Cookies, and other sensitive information are injected through environment variables;
- Real credentials should not be kept in the repository;
- SQLite data is for local or course demonstration environments only;
- All local test credentials and temporary debugging configurations should be removed when deploying publicly.

6. Technical Solution and System Architecture

6.1 Technology Stack

- **Frontend and Display**: Streamlit
- **Data Crawling**: requests
- **Parsing**: BeautifulSoup, lxml
- **Data Processing**: pandas
- **Chinese Word Segmentation**: jieba
- **Sentiment Analysis**: SnowNLP
- **Visualization**: pyecharts, streamlit-echarts, matplotlib, wordcloud
- **Database Storage**: SQLite
- **Optional AI Capability**: DeepSeek API (for keyword filtering)

6.2 System Architecture

```
 User inputs movie name
        |
        ▼
Movie Search Module
        |
        ▼
Review Crawling Module ──────▶ Movie Details Crawling Module
        |
        ▼
```

```
SQLite Database Storage
       |
       ▼
Data Cleaning and Sentiment Analysis Module
       |
       ▼
Visualization Module (Pie Chart / Word Cloud / Timeline / Comparison T
```

6.3 Module Design

- **app.py**: Responsible for page interaction, user input, result display, and progress feedback.
- **crawler/douban_spider.py**: Responsible for searching movies, getting movie details, crawling reviews.
- **analysis/data_cleaner.py**: Responsible for cleaning HTML, links, special characters, and whitespace.
- **analysis/nlp_processor.py**: Responsible for sentiment classification, word frequency statistics, time series, and review weight calculation.
- **database/db_manager.py**: Responsible for database initialization, deduplication insertion, reading movie information and review records.

7. Core Implementation Logic

7.1 Sentiment Analysis Logic

The project adopts a "star rating priority + text supplement" strategy:

- 1-2 stars: Directly considered negative
- 5 stars: Directly considered positive
- 3-4 stars: Combined with text sentiment score for further judgment
- Short texts without star ratings: Default considered neutral

This approach is more suitable for movie short review scenarios than relying solely on sentiment dictionaries, as star ratings themselves already reflect users' clear evaluation tendencies.

7.2 Review Weight Design

To avoid treating ordinary reviews and high-value reviews equally, the system introduces review weights:

- The more extreme the star rating, the higher the weight;
- The higher the like count, the higher the weight;
- The stronger the sentiment intensity, the higher the weight.

This allows simultaneous display of ordinary positive review rate and weighted positive review rate, enhancing the explanatory power of analysis results.

7.3 Keyword Filtering Logic

The project supports two keyword filtering methods:

- **AI Intelligent Filtering**: Prioritize retention of high-frequency words more relevant to movie themes;
- **Regex Fallback Filtering**: When AI is unavailable, filter based on stop words, regex rules, and part-of-speech.

8. System Implementation Results

8.1 Page Function Description

The system currently implements the following main functions:

- Movie name search and candidate movie selection;
- Review crawling progress display;
- Movie rating and total review count display;
- Emotional distribution chart;
- High-frequency word cloud;
- Review popularity timeline;
- Review details sorting display;
- Similar movie sentiment comparison.

8.2 Running Screenshots and Results

Figure 1 Homepage Input Interface

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Figure 1 shows the movie name input area on the system homepage. Users do not need to manually query movie IDs; they only need to input the movie name to start searching, lowering the usage threshold.

Figure 2 Search Results Dropdown Selection Interface

Figure 2 Search Results Dropdown Selection Interface

Figure 2 shows the dropdown selection interface after the system returns candidate movie lists based on user input. Users can directly select the target movie from the results, and the system will automatically bind the corresponding movie ID for subsequent analysis.

Figure 3 Review Crawling Progress Interface

Figure 3 Review Crawling Progress Interface

Figure 3 shows the progress feedback page during review collection. The system displays current crawling page number, current page review count, and cumulative review count in real-time, proving that data is not statically displayed but actually obtained and updated from external sources.

Figure 4 Movie Basic Information and Core Metrics Display (Part 1)

Figure 4 Movie Basic Information and Core Metrics Display 1

Figure 4 shows the system's display of movie basic information and core metrics, including movie name, average rating, total review count, positive review rate, weighted positive review rate, and average sentiment score. This demonstrates that the system performs statistical and value-added processing on raw review data rather than simply displaying raw text.

Figure 5 Movie Basic Information and Core Metrics Display (Part 2)

Figure 5 Movie Basic Information and Core Metrics Display 2

Figure 5 shows the middle result area of the analysis page, which can be used to display emotional distribution charts, keyword visualizations, or other intermediate analysis results. This part shows that the system further transforms review content into graphical information more suitable for user understanding.

Figure 6 Movie Basic Information and Core Metrics Display (Part 3)

Figure 6 Movie Basic Information and Core Metrics Display 3

Figure 6 shows the subsequent result area of the analysis page, which can be used to display review popularity timeline, review detail tables, or similar movie comparisons. Through this part, the system further demonstrates the analysis value of review data in time dimension and comparison dimension.

8.3 Screenshot Summary

The above screenshots cover the main usage workflow of this system:

- User inputs movie name;
- System returns candidate movies for selection;
- System crawls and updates review data;
- System displays statistical indicators and analysis results.

These running results demonstrate that this project can not only crawl external data but also perform cleaning, analysis, and visualization processing on the data, meeting the course assignment requirement of "providing value-added services."

9. Testing and Verification

The project includes multiple test scripts covering the following content:

- Configuration module testing
- Database module testing
- Crawler parsing testing
- Data cleaning testing
- NLP sentiment analysis testing
- Fix verification and regression testing

Completed test materials include:

- `docs/Regression_Test_Report.md`
- `docs/Test_Report_Second_Tester.md`
- `tests/comprehensive_test.py`

These test results indicate that the system has good stability and operability in core functions.

10. Deployment and Accessibility

The current project is built on Streamlit and is suitable for deployment to the following platforms:

- Streamlit Community Cloud
- Render
- Railway

Deployment requirements include:

- Install dependencies from `requirements.txt`;
- Configure environment variables;
- Ensure application entry point is `app.py`;
- Ensure pages display normally across different window sizes.

When formally submitting, the actual accessible public deployment link should be filled in this section.

11. Project Value, Limitations, and Improvement Directions

11.1 Project Value

- Organizes scattered reviews into structured information;
- Helps users quickly understand movie reputation;
- Demonstrates a complete engineering chain from crawling to analysis to display;
- Has potential for further expansion to books, TV dramas, and product review analysis.

11.2 Limitations

- Current data sources mainly rely on a single platform;
- Sentiment analysis is still based on lightweight models and rules, with limited complex semantic processing;
- No automated robots checking implemented;
- Public deployment requires stricter access frequency and anomaly control.

11.3 Future Improvement Directions

- Support more data sources and multi-platform review comparison;
- Use stronger Chinese sentiment models to improve accuracy;

- Add caching and asynchronous crawling to improve performance;
- Add more complete deployment configuration and backend monitoring;
- Add export reports and custom filtering functions.

12. Conclusion

This project completes a Web data service application addressing real-world problems, capable of extracting valuable information from publicly available network data and feeding it back to users in structured, visualized forms. The project has clear problem backgrounds, executable technical routes, and relatively complete function implementations, meeting the course requirements for "crawling external data and providing value-added services."

If real deployment links, demonstration videos, and running screenshots can be supplemented in the future, it will be more suitable for submission as a complete course final project.